



Chapter 23

Microevolution

Lecture Presentations by
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The Smallest Unit of Evolution

- A common misconception is that individual organisms evolve
- Natural selection acts on individuals, but only populations evolve
- Consider, for example, a population of medium ground finches on Daphne Major Island
 - During a drought, large-beaked birds were more likely to crack large seeds and survive

- The finch population evolved, not its individual members

- **Microevolution** is a change in allele frequencies in a population over generations
- Three main mechanisms cause allele frequency change:
 - Natural selection
 - Genetic drift
 - Gene flow
- Only natural selection consistently causes adaptive evolution

Concept 23.1: Genetic variation makes evolution possible

- Variation in heritable traits is a prerequisite for evolution by natural selection
- Gregor Mendel's work on pea plants provided evidence of discrete heritable units (genes)

Genetic Variation

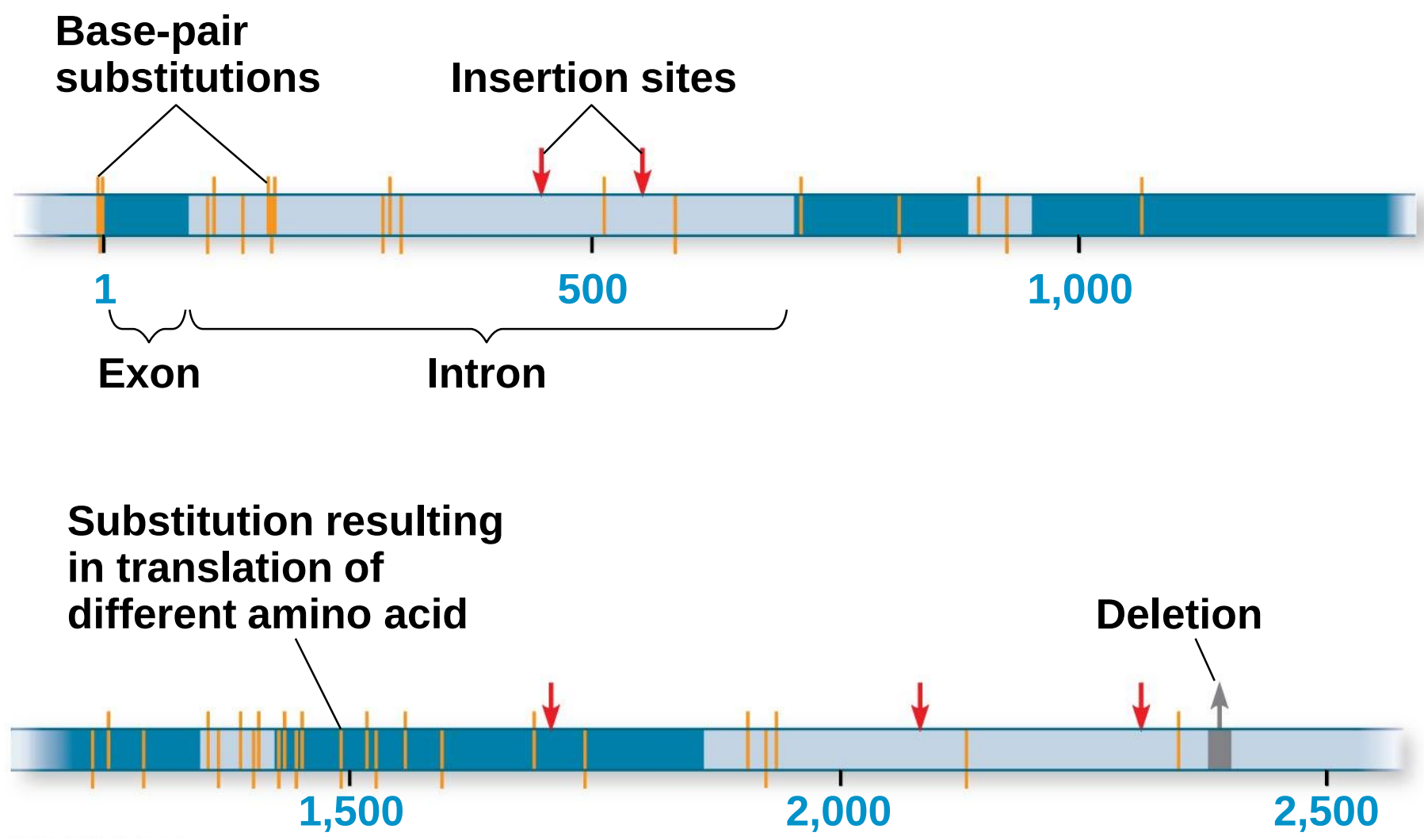
- **Genetic variation** among individuals is caused by differences in genes or other DNA segments
- Phenotype is the product of inherited genotype and environmental influences
- Natural selection can only act on variation with a genetic component

- Some phenotypic differences are determined by a single gene and can be classified on an “either-or” basis
- Other phenotypic differences are determined by the influence of two or more genes and vary along a continuum within a population

- Genetic variation can be measured as gene variability or nucleotide variability
- For gene variability, average heterozygosity measures the average percent of loci that are heterozygous in a population

- Nucleotide variability is measured by comparing the DNA sequences of two or more individuals
- Nucleotide variation rarely results in phenotypic variation
- Most differences occur in noncoding regions (introns)
- Variations that occur in coding regions (exons) rarely change the amino acid sequence of the encoded protein

Figure 23.4



- Some phenotypic variation does not result from genetic differences among individuals, but rather from environmental influences
- Only genetically determined variation can have evolutionary consequences

Sources of Genetic Variation

- New genes and alleles can arise by mutation or gene duplication
- Sexual reproduction can result in genetic variation by recombining existing alleles

Formation of New Alleles

- New alleles arise by mutation, a change in the nucleotide sequence of DNA
- Only mutations in cells that produce gametes can be passed to offspring
- A point mutation is a change in a single nucleotide in a DNA sequence

- The effects of point mutations can vary
 - Mutations that alter the phenotype are often harmful
 - Harmful mutations can be hidden from selection in recessive alleles
 - Mutations that result in a change in phenotype are sometimes, though rarely, beneficial

- The effects of point mutations can vary
 - Point mutations in noncoding regions generally result in **neutral variation**, conferring no selective advantage or disadvantage
 - Mutations to genes can be neutral because of redundancy in the genetic code

Altering Gene Number or Position

- Chromosomal mutations that delete, disrupt, or rearrange many loci are typically harmful
- Duplication of small pieces of DNA increases genome size and is usually less harmful
- Duplicated genes can take on new functions by further mutation
- An ancestral odor-detecting gene has been duplicated many times: Humans have 380 copies of the gene, and mice have 1,200

Rapid Reproduction

- Mutation rates are low in animals and plants
- The average is about one mutation in every 100,000 genes per generation
- Mutation rates are often lower in prokaryotes, but short generation times allow mutations to accumulate rapidly
- Viruses have both high mutation rates and short generation times

Sexual Reproduction

- In organisms that reproduce sexually, most genetic variation results from recombination of alleles
- Sexual reproduction can shuffle existing alleles into new combinations through three mechanisms: crossing over, independent assortment, and fertilization

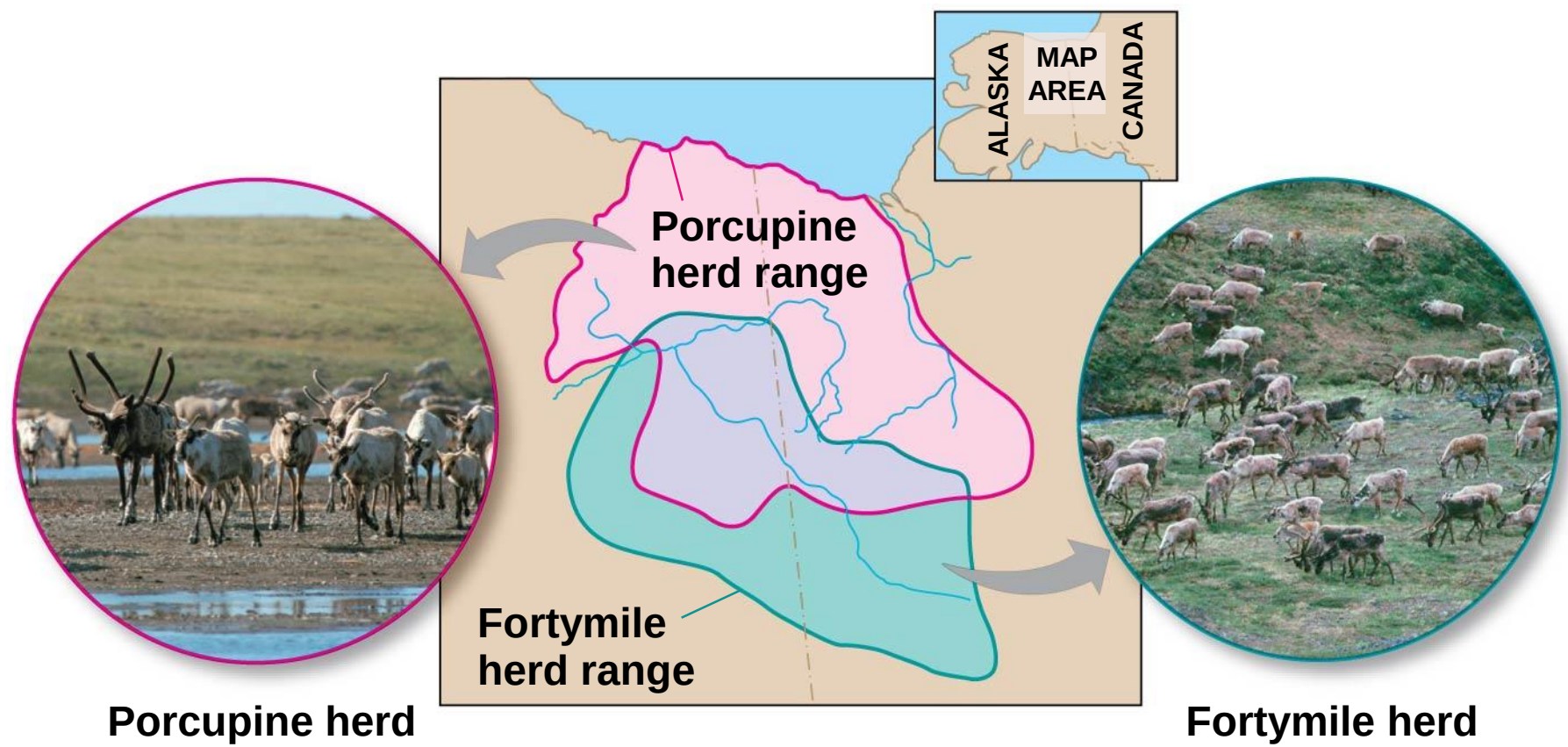
Concept 23.2: The Hardy-Weinberg equation can be used to test whether a population is evolving

- Genetic variation is required for a population to evolve, but does not guarantee that it will
- One or more factors that cause evolution must be at work for a population to evolve

Gene Pools and Allele Frequencies

- A **population** is a localized group of individuals capable of interbreeding and producing fertile offspring
- Populations are not always geographically isolated, but individuals typically only breed with members of their own population

Figure 23.6

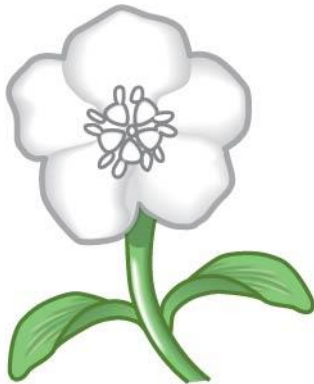


- A **gene pool** consists of all the alleles for all loci in a population
- A locus is fixed if all individuals in a population are homozygous for the same allele
- If there are two or more alleles for a locus, diploid individuals may be either homozygous or heterozygous

Figure 23.8c



$C^R C^R$



$C^W C^W$



$C^R C^W$

- The frequency of an allele in a population can be calculated
 - For diploid organisms, the total number of alleles at a locus is the total number of individuals times 2
 - The total number of dominant alleles at a locus is two alleles for each homozygous dominant individual plus one allele for each heterozygous individual; the same logic applies for recessive alleles

- By convention, if there are two alleles at a locus, p and q are used to represent their frequencies
- The frequency of all alleles in a population will add up to 1
 - That is, $p + q = 1$

- For example, consider a population of wildflowers that is incompletely dominant for color
 - 320 red flowers ($C^R C^R$)
 - 160 pink flowers ($C^R C^W$)
 - 20 white flowers ($C^W C^W$)
- Calculate the number of copies of each allele
 - $C^R = (320 \times 2) + 160 = 800$
 - $C^W = (20 \times 2) + 160 = 200$

- Calculate the frequency of each allele
 - $p = \text{frequency of } C^R = 800/(800 + 200) = 0.8 \text{ (80\%)}$
 - $q = 1 - p = 0.2 \text{ (20\%)}$
- The sum of alleles is always 1 (100%)
 - $0.8 + 0.2 = 1$

The Hardy-Weinberg Equation

- The Hardy-Weinberg equation describes the genetic makeup we expect for a population that is *not* evolving at a particular locus
- If the observed genetic makeup of the population differs from expectations under Hardy-Weinberg, it suggests that the population may be evolving

Hardy-Weinberg Equilibrium

- In a population where gametes contribute to the next generation randomly and Mendelian inheritance occurs, allele and genotype frequencies remain constant from generation to generation
- Such a population is in **Hardy-Weinberg equilibrium**

- Hardy-Weinberg equilibrium describes the constant frequency of alleles in such a gene pool
- Consider, for example, the same population of 500 wildflowers and 1,000 alleles where
 - p = frequency of $C^R = 0.8$
 - q = frequency of $C^W = 0.2$

Figure 23.7

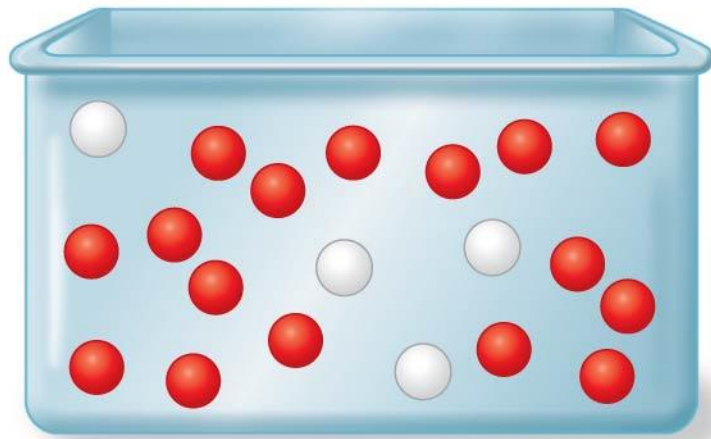
Frequencies of alleles

p = frequency of C^R allele  = 0.8

q = frequency of C^W allele  = 0.2



Alleles in the population



Gametes produced

Each egg:



80%

chance



20%

chance

Each sperm:



80%

chance

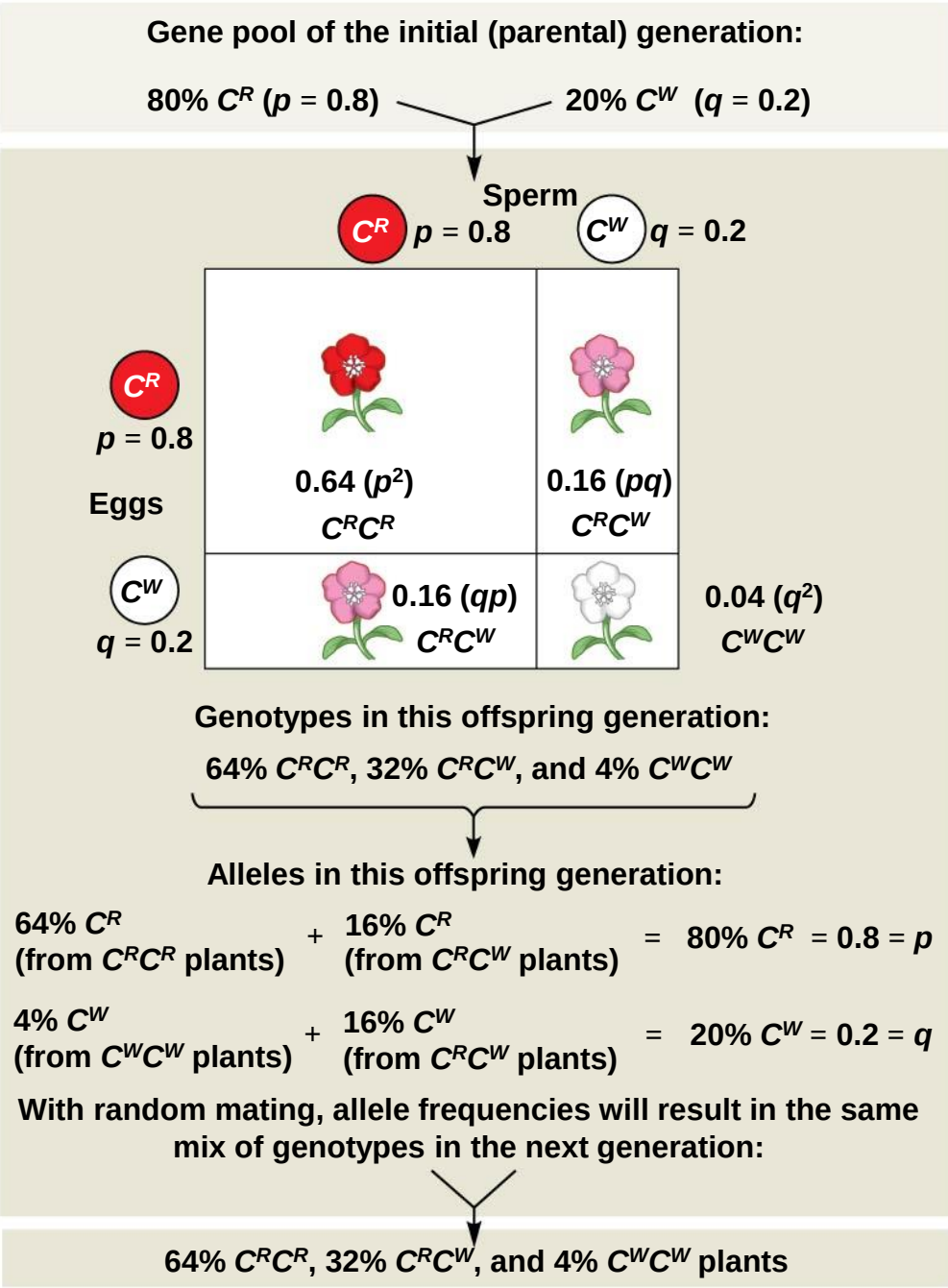


20%

chance

- The frequency of genotypes can be calculated
 - $C^R C^R = p^2 = 0.8 \times 0.8 = 0.64$
 - $C^R C^W = 2pq = 2 \times 0.8 \times 0.2 = 0.32$
 - $C^W C^W = q^2 = 0.2 \times 0.2 = 0.04$
- The frequency of genotypes can be confirmed using a Punnett square

Figure 23.8



- If p and q represent the relative frequencies of the only two possible alleles in a population at a particular locus, then
 - $p^2 + 2pq + q^2 = 1$
 - where p^2 and q^2 represent the frequencies of the homozygous genotypes, and $2pq$ represents the frequency of the heterozygous genotype

Figure 23.8d

$$\begin{array}{ccccccc} p^2 & + & 2pq & + & q^2 & = & 1 \\ \text{Expected} & & \text{Expected} & & \text{Expected} & & \\ \text{frequency} & & \text{frequency} & & \text{frequency} & & \\ \text{of genotype} & & \text{of genotype} & & \text{of genotype} & & \\ C^R C^R & & C^R C^W & & C^W C^W & & \end{array}$$

Conditions for Hardy-Weinberg Equilibrium

- The Hardy-Weinberg approach describes a population that is not evolving
- In real populations, allele and genotype frequencies often do change over time
- Such changes occur when one or more of the conditions for Hardy-Weinberg equilibrium are not met

Table 23.1 Conditions for Hardy-Weinberg Equilibrium

Condition	Consequence if Condition Does Not Hold
1. No mutations	The gene pool is modified if mutations occur or if entire genes are deleted or duplicated.
2. Random mating	If individuals mate within a subset of the population, such as near neighbors or close relatives (inbreeding), random mixing of gametes does not occur and genotype frequencies change.
3. No natural selection	Allele frequencies change when individuals with different genotypes show consistent differences in their survival or reproductive success.
4. Extremely large population size	In small populations, allele frequencies fluctuate by chance over time (a process called genetic drift).
5. No gene flow	By moving alleles into or out of populations, gene flow can alter allele frequencies.

- Natural populations can evolve at some loci while being in Hardy-Weinberg equilibrium at other loci

Applying the Hardy-Weinberg Equation

- The Hardy-Weinberg equation can be used to test whether evolution is occurring in a population
- It is also used to determine the percentage of a population carrying a specific allele

- For example, we can assume that the locus that causes phenylketonuria (PKU) is in Hardy-Weinberg equilibrium given that
 1. The PKU gene mutation rate is low
 2. Mate selection is random with respect to whether or not an individual is a carrier for the PKU allele

3. Natural selection can only act on rare homozygous individuals who do not follow dietary restrictions
4. The population is large
5. Migration has no effect, as many other populations have similar allele frequencies

- The occurrence of PKU is one per 10,000 births
 - $q^2 = 0.0001$
 - $q = 0.01$
- The frequency of normal alleles is
 - $p = 1 - q = 1 - 0.01 = 0.99$
- The frequency of carriers is
 - $2pq = 2 \times 0.99 \times 0.01 = 0.0198$
 - or approximately 2% of the U.S. population

Concept 23.3: Natural selection, genetic drift, and gene flow can alter allele frequencies in a population

- Three major factors alter allele frequencies and bring about most evolutionary change:
 - Natural selection
 - Genetic drift
 - Gene flow

Natural Selection

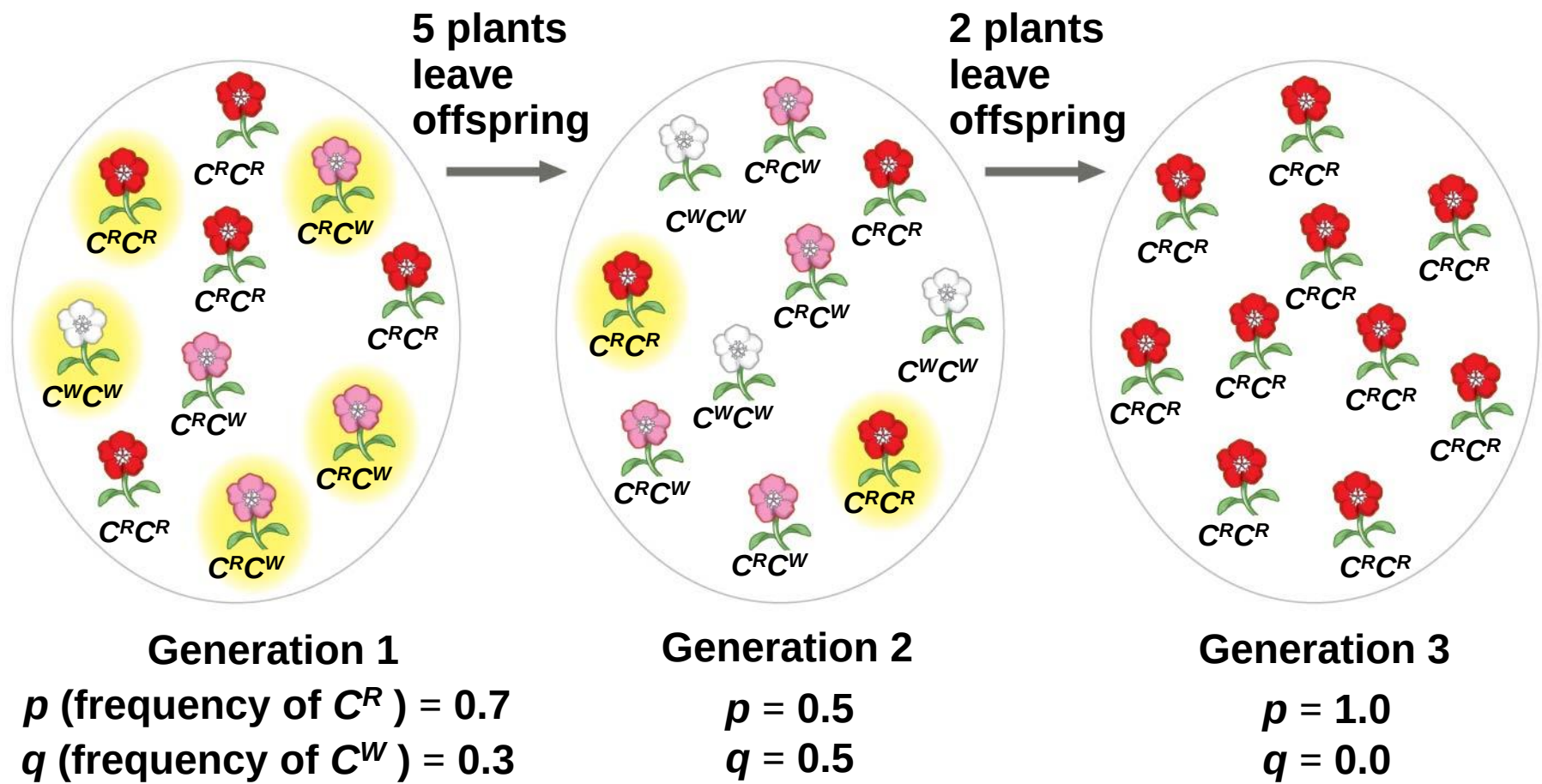
- Selection results in alleles being passed to the next generation in proportions that differ from those in the present generation
- For example, an allele that confers resistance to DDT in fruit flies increased in frequency after DDT was used widely in agriculture

- Natural selection can cause **adaptive evolution**, a process in which traits that enhance survival or reproduction increase in frequency over time

Genetic Drift

- The smaller a sample, the greater the chance of random deviation from a predicted result
- **Genetic drift** describes how allele frequencies fluctuate unpredictably from one generation to the next
- Genetic drift tends to reduce genetic variation through the random loss of alleles

Figure 23.9_3



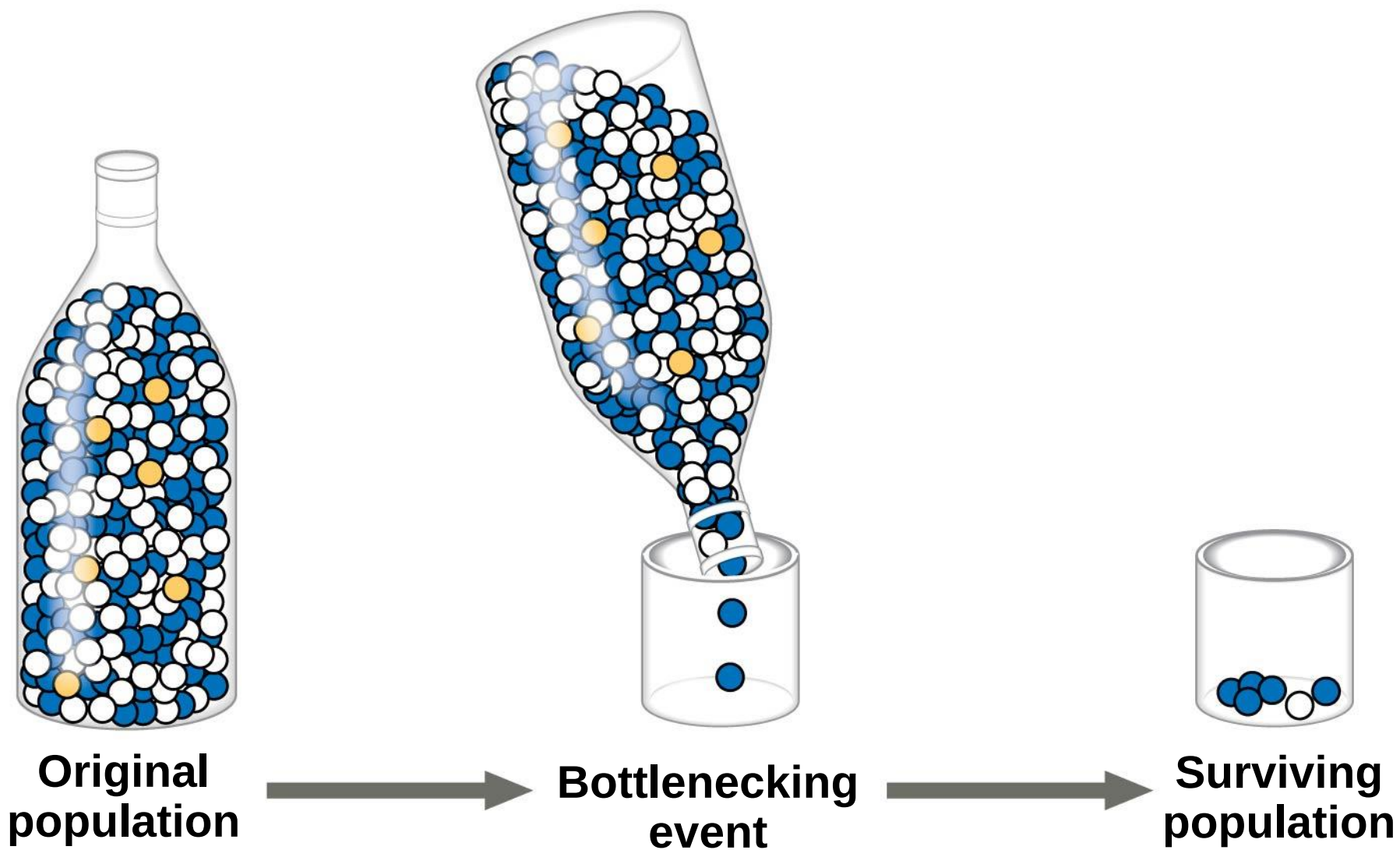
The Founder Effect

- The **founder effect** occurs when a few individuals become isolated from a larger population
- Allele frequencies in the small founder population can be different from those in the larger parent population

The Bottleneck Effect

- The **bottleneck effect** occurs when there is a drastic reduction in population size due to a sudden change in the environment
- The resulting gene pool may no longer be reflective of the original population's gene pool
- If the population remains small, it may be further affected by genetic drift

Figure 23.10_3



- Understanding the bottleneck effect can increase understanding of how human activity affects other species